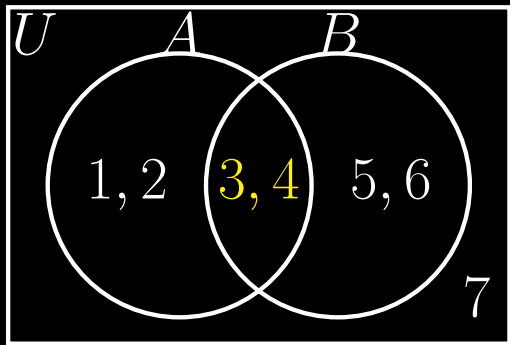


Operations on Sets

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Our Sets

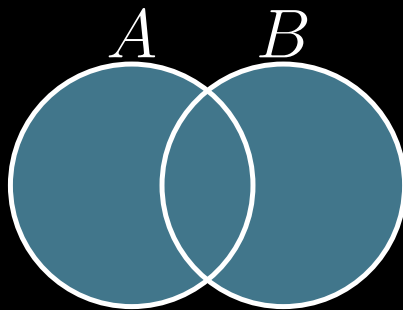


$$U = \{1, 2, 3, 4, 5, 6, 7\}$$

$$A = \{1, 2, 3, 4\} \quad B = \{3, 4, 5, 6\}$$

A **union** of two sets, denoted by $A \cup B$, contains all elements that are in A , in B , or in both.

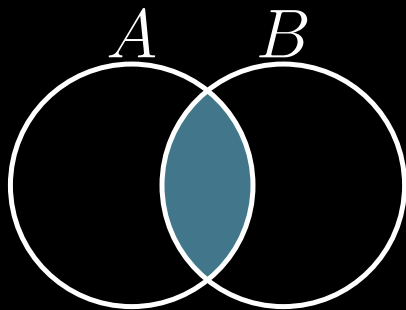
Union



$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

An **intersection** of two sets, denoted by $A \cap B$, contains only the elements common to both sets.

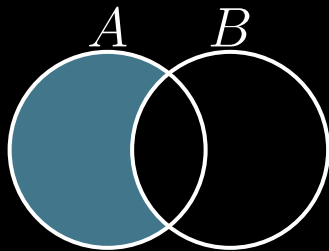
Intersection



$$A \cap B = \{3, 4\}$$

The **difference of sets**, denoted by $A - B$, contains the elements that are in A but not in B .

Difference



$$A - B = \{1, 2\}$$

Order Matters

$A - B$ and $B - A$ are **different**.

$$A - B = \{1, 2\}$$

$$B - A = \{5, 6\}$$

Start with the first set, then remove what is in the second set.

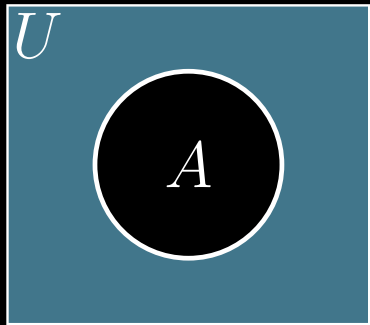
Universal Set

The **universal set** U contains all the elements under consideration.

$$U = \{1, 2, 3, 4, 5, 6, 7\}$$

The **complement** of a set, denoted by A' , contains all elements in the **universal set** that are not in A .

Complement



$$A' = \{5, 6, 7\}$$

Memory Tip

A set and its complement together make up the whole universe.

$$A \cup A' = U$$

A' is **everything** else in *U*.

Summary of Operations

Notation	Meaning	Result
$A \cup B$	in A , in B , or both	$\{1, 2, 3, 4, 5, 6\}$
$A \cap B$	in both A and B	$\{3, 4\}$
$A - B$	in A but not in B	$\{1, 2\}$
A'	in U but not in A	$\{5, 6, 7\}$

Writing the Answer

- ▶ Always list the elements of a set only once.
- ▶ Arrange the elements in a consistent order when writing the resulting set.

Examples

Example 1

Let $A = \{2, 3, 4, 5, 9\}$, $B = \{4, 5, 6, 7\}$. Find $A \cup B$, $A \cap B$, $A - B$, and $B - A$.

$$A \cup B = \{2, 3, 4, 5, 6, 7, 9\}$$

$$A \cap B = \{4, 5\}$$

$$A - B = \{2, 3, 9\} \quad B - A = \{6, 7\}$$

Example 2

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{2, 3, 5, 7\}$. Find A' .

Elements of U that are not in A :

$$A' = \{1, 4, 6, 8\}$$

Example 3

Let $A = \{a, b, c, d\}$, $B = \{c, d, e\}$. Find $A \cup B$, $A \cap B$, $A - B$, and $B - A$.

$$A \cup B = \{a, b, c, d, e\}$$

$$A \cap B = \{c, d\}$$

$$A - B = \{a, b\} \quad B - A = \{e\}$$

Example 4

Let $U = \{p, q, r, s, t, u\}$, $C = \{q, s, u\}$. Find C' and $C \cup C'$.

$$C' = \{p, r, t\}$$

$$C \cup C' = \{p, q, r, s, t, u\} = U$$

Example 5

In a class survey, $S = \{\text{Rey, Sam, Tina, Uma}\}$ like math and $T = \{\text{Sam, Uma, Vic}\}$ like science. Find $S \cup T$, $S \cap T$, and $S - T$.

$$S \cup T = \{\text{Rey, Sam, Tina, Uma, Vic}\}$$

$$S \cap T = \{\text{Sam, Uma}\}$$

$$S - T = \{\text{Rey, Tina}\}$$

Practice Exercises

Practice 1

Let $A = \{1, 2, 3, 5, 7\}$, $B = \{2, 4, 5, 6, 8\}$.

Find $A \cup B$, $A \cap B$, $A - B$, and $B - A$.

Practice 2

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 4, 6, 8, 10\}$. Find A' .

Practice 3

Let $A = \{a, b, c, d, e\}$, $B = \{c, d, e, f, g\}$.

Find $A \cup B$, $A \cap B$, and $A - B$.

Practice 4

Let $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4, 5\}$. Find $(A \cup B)$ and $(A \cap B)$.

Practice 5

Let $U = \{a, b, c, d, e, f, g, h\}$, $C = \{b, d, f, h\}$. Find C' .

Practice 6

Let $M = \{3, 6, 9, 12, 15\}$, $N = \{6, 12, 18, 24\}$. Find $M \cup N$, $M \cap N$, and $M - N$.

Practice 7

Let $P = \{x, y, z\}$, $Q = \{y, z, w, v\}$. Find $P \cup Q$, $P \cap Q$, and $Q - P$.

Practice 8

Let $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $R = \{1, 3, 5, 7, 9\}$. Find R' and $R \cup \{0\}$.

Practice 9

Let $A = \{2, 3, 5, 7, 11\}$, $B = \{5, 7, 13, 17\}$.

Find $A \cup B$, $A \cap B$, and $B - A$.

Activities

Activity 1

Let $A = \{3, 5, 7, 9, 11\}$, $B = \{1, 3, 5, 8, 10\}$.

Find $A \cup B$, $A \cap B$, $A - B$, and $B - A$.

Activity 2

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$, $A = \{1, 3, 5, 7, 9, 11\}$. Find A' .

Activity 3

Let $A = \{m, n, o, p, q\}$, $B = \{o, p, q, r, s\}$.

Find $A \cup B$, $A \cap B$, and $A - B$.

Activity 4

Let $A = \{5, 10, 15, 20\}$, $B = \{10, 15, 25, 30\}$.
Find $(A \cup B)$ and $(A \cap B)$.

Activity 5

Let $U = \{a, b, c, d, e, f, g, h, i, j\}$, $C = \{a, c, e, g, i\}$. Find C' .

Activity 6

Let $M = \{4, 8, 12, 16, 20\}$, $N = \{8, 16, 24, 32\}$. Find $M \cup N$, $M \cap N$, and $M - N$.

Activity 7

Let $P = \{k, l, m\}$, $Q = \{m, n, o, l\}$. Find $P \cup Q$, $P \cap Q$, and $Q - P$.

Activity 8

Let $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $R = \{0, 2, 4, 6, 8\}$. Find R' and $R \cup \{9\}$.

Thank you!